

Linux System & Network Administration Lab

Project Documentation

A record of hands-on system and network administration work performed on a self-provisioned Ubuntu virtual machine

Overview

This project was built to gain practical, hands-on experience with core Linux system and network administration tasks rather than studying them in the abstract. I provisioned my own Ubuntu virtual machine and worked through user and permission management, service administration, package management, networking, security hardening, and log analysis, executing every command directly. Where issues arose during the process, the troubleshooting steps are documented alongside the resolution, since diagnosing and resolving real problems was a core part of the exercise.

Environment

- Operating system: Ubuntu 26.04 LTS Desktop
- Hypervisor: Oracle VirtualBox, on a Windows host
- Networking: VirtualBox NAT network (10.0.2.0/24)
- VirtualBox Guest Additions installed for host/guest clipboard integration

Initial VM provisioning and the Ubuntu installation itself are not covered in this document, as they follow the standard installation process and do not reflect administration skills. This write-up begins at the point where hands-on administration work started.

1. Filesystem, Permissions & Users

This section covers the foundational layer that everything else depends on: file ownership, access control, and how user accounts and group membership are structured.

Command	Purpose
<code>ls -la</code>	List files with permissions, ownership, and hidden files included
<code>chmod 700 <dir></code>	Restrict a directory to owner-only access, using octal notation (read=4, write=2, execute=1 per digit)
<code>useradd -m <user></code>	Create a new user account, with -m also creating that user's home directory
<code>passwd <user></code>	Set or change a user's password
<code>usermod -aG sudo <user></code>	Add a user to the sudo group, appending rather than overwriting existing group memberships

This section also included reviewing the field structure of `/etc/passwd` and `/etc/group`, and confirming the distinction between `sudo <command>`, which authenticates and elevates the invoking user for a single command, and switching to a persistent root shell — a distinction relevant to maintaining least-privilege access.

```

vboxuser@ubuntu-admin-labs:~$ ls -la ~ | grep Documents
vboxuser@ubuntu-admin-labs:~$ ls -la ~ | grep Documents
vboxuser@ubuntu-admin-labs:~$ ls -la ~ | grep Documents
drwx----- 2 vboxuser vboxuser 4096 Aug 23 07:05 Documents
vboxuser@ubuntu-admin-labs:~$

```

SS/01-filesystem-permissions-chmod-verified.png — chmod 700 applied to ~/Documents, verified with ls -la showing drwx-----

```

vboxuser@ubuntu-admin-labs:~$ id alex
uid=1001(alex) gid=1001(alex) groups=1001(alex)
vboxuser@ubuntu-admin-labs:~$ sudo usermod -aG sudo alex
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
vboxuser@ubuntu-admin-labs:~$ id alex
uid=1001(alex) gid=1001(alex) groups=1001(alex),27(sudo)
vboxuser@ubuntu-admin-labs:~$ grep alex /etc/passwd
alex:x:1001:1001::/home/alex:/bin/sh
vboxuser@ubuntu-admin-labs:~$ grep sudo /etc/group
grep: /etc/group: No such file or directory
vboxuser@ubuntu-admin-labs:~$ grep sudo /etc/group
sudo:x:27:vboxuser,alex
vboxuser@ubuntu-admin-labs:~$

```

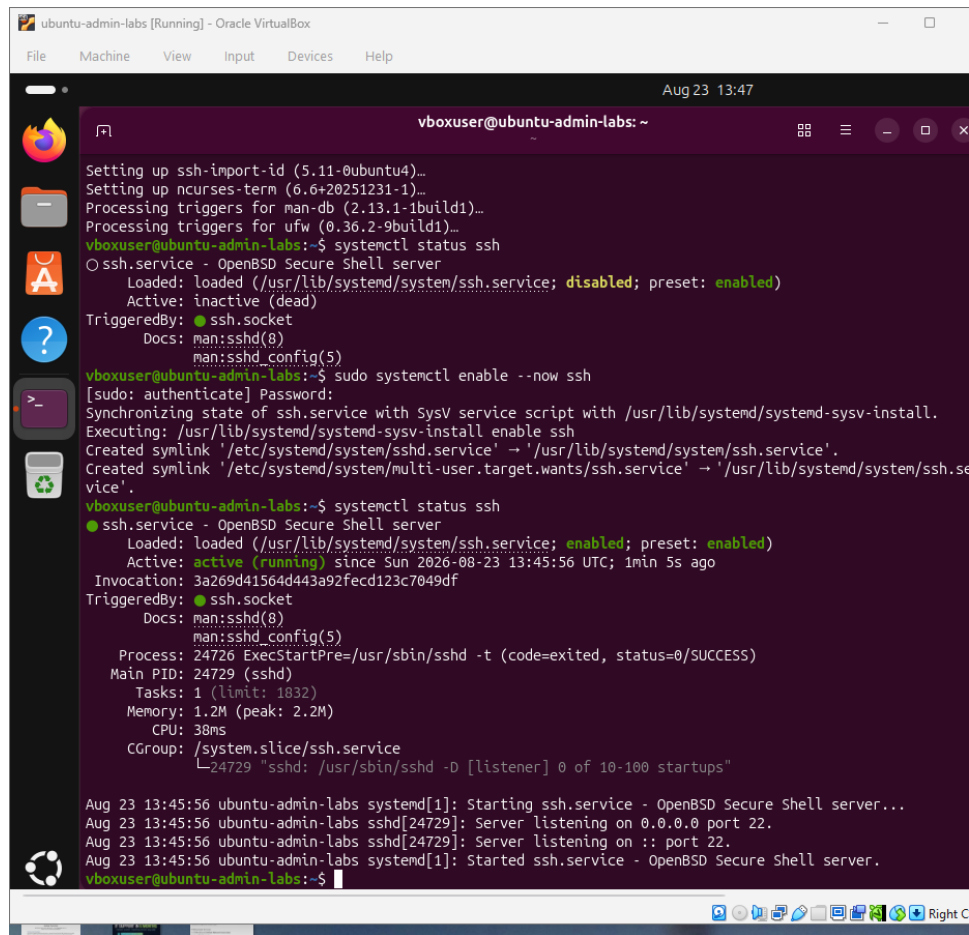
SS/01-filesystem-permissions-user-group-sudo.png — Confirming usermod -aG sudo succeeded via id and /etc/group, and identifying the user's default shell

2. Processes & Services

This section covers inspecting what is currently running on the system, controlling long-running background services, and scheduling recurring tasks.

Command	Purpose
ps aux	Snapshot of every running process on the system
top	Live view of process activity, load average, and CPU/memory usage
systemctl status / enable --now <service>	Check a service's status, and start it immediately while also enabling it to start automatically on every future boot
journalctl -xe	View recent system log entries with added explanation
crontab -e	Edit a user's scheduled jobs, using the five-field time syntax: minute, hour, day-of-month, month, day-of-week

To confirm the cron job was functioning correctly rather than assuming it based on syntax alone, a test job was configured to append a timestamped entry to a log file every minute, with the output verified directly. A stopped background job was also cleared using kill %1, reinforcing the distinction between a shell job number (%1) and a process ID.



```
ubuntu-admin-labs [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-admin-labs: ~
Aug 23 13:47

Setting up ssh-import-id (5.11-0ubuntu4)...
Setting up ncurses-term (6.6+20251231-1)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for ufw (0.36.2-9build1)...
vboxuser@ubuntu-admin-labs:~$ systemctl status ssh
○ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
     TriggeredBy: ● ssh.socket
                Docs: man:sshd(8)
                   man:sshd_config(5)
vboxuser@ubuntu-admin-labs:~$ sudo systemctl enable --now ssh
[sudo: authenticate] Password:
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
vboxuser@ubuntu-admin-labs:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-08-23 13:45:56 UTC; 1min 5s ago
     Invocation: 3a269d41564d443a92fecdd123c7049df
     TriggeredBy: ● ssh.socket
                Docs: man:sshd(8)
                   man:sshd_config(5)
   Process: 24726 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 24729 (sshd)
       Tasks: 1 (limit: 1832)
      Memory: 1.2M (peak: 2.2M)
         CPU: 38ms
    CGroup: /system.slice/ssh.service
            └─24729 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 23 13:45:56 ubuntu-admin-labs systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 23 13:45:56 ubuntu-admin-labs sshd[24729]: Server listening on 0.0.0.0 port 22.
Aug 23 13:45:56 ubuntu-admin-labs sshd[24729]: Server listening on :: port 22.
Aug 23 13:45:56 ubuntu-admin-labs systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
vboxuser@ubuntu-admin-labs:~$
```

SS/02-processes-services-ssh-active.png — systemctl status ssh after enable --now, confirming the service is active and listening on port 22

```
Aug 25 11:14
vboxuser@ubuntu-admin-labs: ~

vboxuser@ubuntu-admin-labs:~$ crontab -e
crontab: installing new crontab
vboxuser@ubuntu-admin-labs:~$ crontab -e
No modification made
vboxuser@ubuntu-admin-labs:~$ crontab -i
crontab: usage error: file name must be specified for replace
usage: crontab [-u user] [-n] file
       crontab [-u user] [-i] { -e | -l | -r }

       -h      (displays this help message)

       file    (default operation is replace, per 1003.2)
       -n      (dry run: checks the syntax, then bails out)
       -u user  (choose the user whose crontab is touched)

       -e      (edit user's crontab)
       -l      (list user's crontab)
       -r      (delete user's crontab)

       -i      (prompt before deleting user's crontab)
vboxuser@ubuntu-admin-labs:~$
```

SS/02-processes-cron-crontab-l-verified.png — crontab -l showing the final, correctly saved cron job line

```
-l      (list user's crontab)
-r      (delete user's crontab)

-i      (prompt before deleting user's crontab)
vboxuser@ubuntu-admin-labs:~$ crontab -l

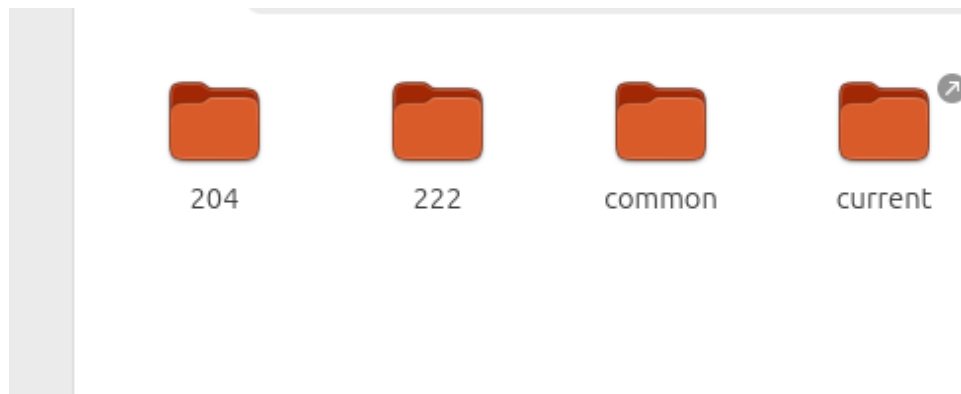
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
* * * * * echo "cron works: $(date)" >> /home/vboxuser/cron-test.log
vboxuser@ubuntu-admin-labs:~$
```

SS/02-processes-cron-test-log-fired.png — cat ~/cron-test.log showing two timestamped entries one minute apart, confirming the job fired on schedule

3. Package Management

Ubuntu's package management is structured in two layers: apt, the high-level tool used for day-to-day installing, updating, and searching; and dpkg, the lower-level tool underneath it, used here primarily for inspecting what is already installed.

Command	Purpose
<code>sudo apt update && sudo apt full-upgrade -y</code>	Refresh the package index, then non-interactively upgrade every package, allowing apt to add or remove packages as needed to satisfy dependency changes
<code>sudo apt install <package> -y</code>	Install a package without an interactive confirmation prompt
<code>apt-cache search <term></code>	Find packages by matching a keyword against package names and descriptions
<code>dpkg -l grep <package></code>	List installed packages, filtered to confirm a specific one is present



SS/03-package-management-tree-install.png — Installing the tree package and using it to display the home directory as a visual tree, including Snap's versioned package layout

```
ii bc 1.07.1-4build1 amd64 GNU B
ii bind9-dnsutils 1:9.20.24-1ubuntu0.2 amd64 Clie
ii bind9-host 1:9.20.24-1ubuntu0.2 amd64 DNS L
ii bind9-libs:amd64 1:9.20.24-1ubuntu0.2 amd64 Share

[1]+  Stopped                  dpkg -l
vboxuser@ubuntu-admin-labs:~$ dpkg -l |grep tree
ii libhtml-tree-perl 5.07-3 all Perl
odule to represent and create HTML syntax trees
ii libxml-twig-perl 1:3.54-1build1 all Perl
odule for processing huge XML documents in tree mode
ii libxml-xpathengine-perl 0.14-2 all re-us
ble XPath engine for DOM-like trees
ii tree 2.3.1-1 amd64 displ
ys an indented directory tree, in color
vboxuser@ubuntu-admin-labs:~$
```

SS/03-package-management-dpkg-verified.png — dpkg -l piped through grep, confirming the tree package is installed with its version and description

4. Core Networking

This section covers addressing, routing, and name resolution — the fundamentals of how the machine communicates on a network.

Command	Purpose
<code>ip a</code>	Show every network interface and its assigned IP address
<code>ip route</code>	Show the routing table, including the default gateway
<code>nmcli device status</code>	NetworkManager's view of active network connections
<code>ping -c 4 <host></code>	Test basic reachability with a fixed number of packets
<code>dig <domain></code>	Resolve a domain name and inspect the full DNS response
<code>ss -tulpn</code>	List listening TCP and UDP ports along with the owning process

A key distinction established in this section is between a service bound to 0.0.0.0, which is reachable from other machines on the network, and one bound to 127.0.0.1 (loopback), which is reachable only from processes running on the same machine — a distinction directly relevant to the firewall configuration covered later.

```
vboxuser@ubuntu-admin-labs:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:63:55:76 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83513sec preferred_lft 83513sec
    inet6 fd17:625c:f037:2:a00:27ff:fe63:5576/64 scope global dynamic noprefixroute
        valid_lft 86369sec preferred_lft 14369sec
    inet6 fe80::a00:27ff:fe63:5576/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
vboxuser@ubuntu-admin-labs:~$ s
```

SS/04-networking-ip-a-interface.png — `ip a` showing the `enp0s3` interface with its assigned IP address on the VirtualBox NAT network

```
lo    loopback    connected (externally)  lo
vboxuser@ubuntu-admin-labs:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=47.8 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=50.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=26.3 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=52.5 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 26.284/44.272/52.455/10.515 ms
```

SS/04-networking-ping-8888.png — `ping -c 4 8.8.8.8` completing with 0% packet loss, confirming outbound connectivity

```
vboxuser@ubuntu-admin-labs:~$ dig google.com

; <<>> DiG 9.20.24-1ubuntu0.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 39469
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                267     IN      A      142.250.195.110

;; Query time: 31 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Aug 25 12:10:49 UTC 2026
;; MSG SIZE  rcvd: 55

vboxuser@ubuntu-admin-labs:~$
```

SS/04-networking-dig-google.png — dig google.com, with the ANSWER SECTION showing the resolved A record

```
top  LISTEN 0      4096      *:*          *:*          *:*
vboxuser@ubuntu-admin-labs:~$ sudo ss -tulpn
[sudo: authenticate] Password:
Netid  State Recv-Q  Send-Q      Local Address:Port      Peer Address:Port      Process
tcp    LISTEN 0      4096      0.0.0.0:22      0.0.0.0:*        users:((("sshd",pid=1214,fid=3),("systemd",pid=1,fid=403)))
tcp    LISTEN 0      4096      127.0.0.1:8081  0.0.0.0:*        users:((("cupsd",pid=1153,fid=7)))
tcp    LISTEN 0      4096      127.0.0.53:53   0.0.0.0:*        users:((("systemd-resolve",pid=381,fid=17),("systemd-resolve",pid=381,fid=19)))
tcp    LISTEN 0      4096      :::22          :::*             users:((("sshd",pid=1214,fid=4),("systemd",pid=1,fid=403)))
tcp    LISTEN 0      4096      :::8081        :::*             users:((("cupsd",pid=1153,fid=6)))
udp    UNCONN 0        0          0.0.0.0:5353    0.0.0.0:*        users:((("avahi-daemon",pid=780,fid=12)))
udp    UNCONN 0        0          10.0.2.15:3702  0.0.0.0:*        users:((("python3",pid=5009,fid=10)))
udp    UNCONN 0        0          259.255.255.255:3702  0.0.0.0:*        users:((("python3",pid=5009,fid=11)))
udp    UNCONN 0        0          127.0.0.54:53   0.0.0.0:*        users:((("systemd-resolve",pid=381,fid=18)))
udp    UNCONN 0        0          127.0.0.53:53   0.0.0.0:*        users:((("systemd-resolve",pid=381,fid=16)))
udp    UNCONN 0        0          127.0.0.1:323   0.0.0.0:*        users:((("chronyd",pid=942,fid=4)))
udp    UNCONN 0        0          0.0.0.0:33721   0.0.0.0:*        users:((("python3",pid=5009,fid=9)))
udp    UNCONN 0        0          *:52261         *:~              users:((("python3",pid=5009,fid=12)))
udp    UNCONN 0        0          :::5353         :::*             users:((("avahi-daemon",pid=780,fid=13)))
udp    UNCONN 0        0          [fe80::a00:27ff:fe63:5570]:3702  :::*             users:((("python3",pid=5009,fid=13)))
udp    UNCONN 0        0          [ff02::c]:3702  :::*             users:((("python3",pid=5009,fid=11)))
udp    UNCONN 0        0          :::323          :::*             users:((("chronyd",pid=942,fid=5)))
tcp    LISTEN 0      4096      0.0.0.0:22      0.0.0.0:*        users:((("sshd",pid=1214,fid=3),("systemd",pid=1,fid=402)))
tcp    LISTEN 0      4096      127.0.0.1:8081  0.0.0.0:*        users:((("cupsd",pid=1153,fid=7)))
tcp    LISTEN 0      4096      127.0.0.53:53   0.0.0.0:*        users:((("systemd-resolve",pid=381,fid=17),("systemd-resolve",pid=381,fid=19)))
tcp    LISTEN 0      4096      :::22          :::*             users:((("sshd",pid=1214,fid=4),("systemd",pid=1,fid=403)))
tcp    LISTEN 0      4096      :::8081        :::*             users:((("cupsd",pid=1153,fid=6)))
```

SS/04-networking-ss-tulpn-sudo.png — sudo ss -tulpn showing listening ports with full process names (sshd, cupsd, systemd-resolve)

5. Diagnostics & Recon

This section shifts from reviewing local machine configuration to actively probing the network: discovering live hosts, scanning for open ports and running services, and capturing raw traffic.

Command	Purpose
sudo nmap -sn <subnet>	Host discovery — sweep a subnet to identify which addresses are currently in use, without scanning ports

<code>sudo nmap -sV -p- <host></code>	Full port scan across all 65,535 ports, with service and version fingerprinting on anything open
<code>sudo tcpdump -i <interface> -c <count></code>	Capture a fixed number of live packets on a given network interface
<code>nc -zv <host> <port></code>	Test whether a specific port accepts a connection, without sending any data

```

Processing triggers for libc-bin (2.43-2ubuntu2.5)...
vboxuser@ubuntu-admin-labs:~$ sudo nmap -sn 10.0.2.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-25 12:22 +0000
Nmap scan report for _gateway (10.0.2.2)
Host is up (0.00055s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.2.3
Host is up (0.00038s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for ubuntu-admin-labs.local (10.0.2.15)
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.29 seconds
vboxuser@ubuntu-admin-labs:~$

```

SS/05-diagnostics-nmap-host-discovery.png — `sudo nmap -sn` across the VirtualBox NAT subnet, identifying the gateway, the DHCP helper, and the VM itself

```

Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.29 seconds
vboxuser@ubuntu-admin-labs:~$ sudo nmap -sV -p- 10.0.2.15
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-25 12:24 +0000
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for ubuntu-admin-labs.local (10.0.2.15)
Host is up (0.000037s latency).
Not shown: 65534 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.2p1 Ubuntu 2ubuntu3.5 (Ubuntu Linux; protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

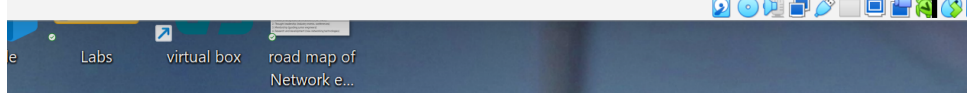
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 109.38 seconds
vboxuser@ubuntu-admin-labs:~$

```

SS/05-diagnostics-nmap-port-scan-ssh.png — Full 65,535-port scan completing with only port 22 open, correctly fingerprinted as OpenSSH


```
vboxuser@ubuntu-admin-labs:~$ sudo tcpdump -i enp0s3 -c 10
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), snapshot length 262144 bytes

12:28:46.077714 IP ubuntu-admin-labs.36557 > time.cloudflare.com.ntp: NTPv4, Client, le
12:28:46.080795 IP ubuntu-admin-labs.55631 > 10.94.100.86.domain: 38054+ PTR? 123.200.1
addr.arpa. (46)
12:28:46.170464 IP time.cloudflare.com.ntp > ubuntu-admin-labs.36557: NTPv4, Server, le
12:28:46.170464 IP 10.94.100.86.domain > ubuntu-admin-labs.55631: 38054 1/0/0 PTR time.
.com. (79)
12:28:46.172260 IP ubuntu-admin-labs.42450 > 10.94.100.86.domain: 29812+ PTR? 15.2.0.10
rpa. (40)
12:28:46.191589 IP 10.94.100.86.domain > ubuntu-admin-labs.42450: 29812 NXDomain 0/0/0
12:28:46.193070 IP ubuntu-admin-labs.44525 > 10.94.100.86.domain: 9089+ PTR? 86.100.94.
.arpa. (43)
12:28:46.255536 IP 10.94.100.86.domain > ubuntu-admin-labs.44525: 9089 NXDomain 0/1/0 (
12:28:51.309568 IP ubuntu-admin-labs.52732 > 193.177.220.128.ntp: NTPv4, Client, length
12:28:51.346541 ARP, Request who-has _gateway tell ubuntu-admin-labs, length 28
10 packets captured
16 packets received by filter
0 packets dropped by kernel
vboxuser@ubuntu-admin-labs:~$
```



SS/05-diagnostics-tcpdump-capture.png — tcpdump capturing live NTP, DNS, and ARP traffic on the primary network interface

6. Firewalls & Remote Access

This section required particular care: an incorrect order of operations here can result in losing remote access to the machine entirely.

Command	Purpose
<code>sudo ufw allow ssh</code>	Explicitly allow SSH before enabling the firewall, so remote access is not accidentally blocked
<code>sudo ufw enable</code>	Switch the firewall to a default-deny-incoming policy, with outgoing traffic still allowed
<code>sudo ufw status verbose</code>	Review the active policy and every configured rule
<code>ssh <user>@<host></code>	Open a remote shell session over an encrypted connection
<code>ssh-keygen -t ed25519</code>	Generate a new public/private key pair for key-based authentication
<code>ssh-copy-id <user>@<host></code>	Install a public key into the target account's <code>authorized_keys</code> file
<code>/etc/ssh/sshd_config</code>	The server-side SSH policy file, controlling settings such as root login and password authentication
<code>sudo apt install fail2ban -y</code>	Install a service that automatically bans IP addresses after repeated failed login attempts

SSH was explicitly allowed before enabling the firewall, since reversing that order would have removed the access needed to correct any misconfiguration. Key-based authentication was then configured and verified two ways: a login that required no password prompt at all, and confirmation in the authentication log showing "Accepted publickey" rather than "Accepted password."

```
3 sessions listed.
vboxuser@ubuntu-admin-labs:~$ loginctl list-sessions
SESSION UID USER SEAT LEADER CLASS TTY IDLE SINCE
2 1000 vboxuser seat0 2546 user tty2 no -
3 1000 vboxuser - 2609 manager - no -
99 1000 vboxuser - 8532 user - no -

3 sessions listed.
vboxuser@ubuntu-admin-labs:~$ exit
logout
Connection to 10.0.2.15 closed.
vboxuser@ubuntu-admin-labs:~$ sudo ufw allow ssh
[sudo: authenticate] Password:
Rules updated
Rules updated (v6)
vboxuser@ubuntu-admin-labs:~$ sudo ufw enable
Firewall is active and enabled on system startup
vboxuser@ubuntu-admin-labs:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)

vboxuser@ubuntu-admin-labs:~$
```

SS/06-firewall-ufw-enabled-sessions.png — ufw allow ssh, enable, and status verbose confirming the firewall is active with only port 22 allowed; loginctl list-sessions distinguishing the console login from the SSH session

```
Status: inactive
vboxuser@ubuntu-admin-labs:~$ ssh vboxuser@10.0.2.15
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.
ED25519 key fingerprint is: SHA256:GACzyDM0A8KgfZwFr+6kL+zMx4W5Ir5cl6pUxeWLFHM
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? -y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '10.0.2.15' (ED25519) to the list of known hosts.
vboxuser@10.0.2.15's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

vboxuser@ubuntu-admin-labs:~$
```

SS/06-firewall-ssh-login-fingerprint.png — First SSH login: host key fingerprint verification, followed by a successful login and the Ubuntu welcome banner

```
ubuntu-admin-labs [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Aug 26 06:53
vboxuser@ubuntu-admin-labs: ~ — ssh vboxuser@10.0.2.15
* Documentation: https://docs.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
Last login: Tue Aug 25 21:28:37 2026 from 10.0.2.15
vboxuser@ubuntu-admin-labs:~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/vboxuser/.ssh/id_ed25519):
Enter passphrase for "/home/vboxuser/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/vboxuser/.ssh/id_ed25519
Your public key has been saved in /home/vboxuser/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:kwhegKedJsEp/CF8rGsOEi7hvFrgbyKo3BNawkt9JkI vboxuser@ubuntu-admin-labs
The key's randomart image is:
+--[ED25519 256]--+
|Eoo.          |
|oO=*.+       |
|+.B=*.       |
|O+OO+O . .   |
|=ooo. . S    |
|=B .         |
|B.o .        |
|+=oo         |
|*.oo.        |
+----[SHA256]-----+
vboxuser@ubuntu-admin-labs:~$ ssh-copy-id vboxuser@10.0.2.15
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/vboxuser/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install
new keys
vboxuser@10.0.2.15's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'vboxuser@10.0.2.15'"
and check to make sure that only the key(s) you wanted were added.
vboxuser@ubuntu-admin-labs:~$
```

SS/06-firewall-ssh-keygen-ed25519.png — ssh-keygen -t ed25519 generating a new key pair, including its randomart fingerprint image

```
Aug 26 06:54
vboxuser@ubuntu-admin-labs: ~ — ssh vboxuser@10.0.2.15

Enter passphrase for "/home/vboxuser/.ssh/id_ed25519" (empty for no passphrase)
Enter same passphrase again:
Your identification has been saved in /home/vboxuser/.ssh/id_ed25519
Your public key has been saved in /home/vboxuser/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:kwhegKedJsEp/CF8rGsOEi7hvFrgbyKo3BNAwkt9JkI vboxuser@ubuntu-admin-labs
The key's randomart image is:
+--[ED25519 256]--+
|Eoo.              |
|oO=*+.            |
|.B=*..            |
|o+oo+o . .        |
|=ooo. . S         |
|=B .              |
|B.o .             |
|+=oo              |
|*.oo.             |
+----[SHA256]-----+
vboxuser@ubuntu-admin-labs:~$ ssh-copy-id vboxuser@10.0.2.15
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/vboxuser/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted, it
vboxuser@10.0.2.15's password:
Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'vboxuser@10.0.2.15'"
and check to make sure that only the key(s) you wanted were added.

vboxuser@ubuntu-admin-labs:~$ ssh vboxuser@10.0.2.15
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Aug 26 06:48:06 2026 from 10.0.2.15
vboxuser@ubuntu-admin-labs:~$
```

SS/06-firewall-ssh-copy-id-passwordless.png — ssh-copy-id installing the public key, immediately followed by an SSH login with no password prompt at all

```
Aug 26 07:19
vboxuser@ubuntu-admin-labs: ~ — ssh vboxuser@10.0.2.15

[sudo: authenticate] Password:
#PermitRootLogin prohibit-password
#PasswordAuthentication yes
# PasswordAuthentication. Depending on your PAM configuration,
# the setting of "PermitRootLogin prohibit-password".
# PAM authentication, then enable this but set PasswordAuthentication
vboxuser@ubuntu-admin-labs:~$ sudo apt install fail2ban -y
sudo systemctl status fail2ban
Installing:
  fail2ban

Installing dependencies:
  python3-pyasyncore python3-pyinotify whois

Suggested packages:
  mailx monit sqlite3 python-pyinotify-doc

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 6
  Download size: 510 kB
  Space needed: 2,620 kB / 17.9 GB available

Get:1 http://au.archive.ubuntu.com/ubuntu resolute/universe amd64 fail2ban all 1.1.0-9 [421 kB]
Get:2 http://au.archive.ubuntu.com/ubuntu resolute/main amd64 python3-pyasyncore all 1.0.2-3build1 [10.4 kB]
Get:3 http://au.archive.ubuntu.com/ubuntu resolute/main amd64 python3-pyinotify all 0.9.6-5build1 [25.5 kB]
Get:4 http://au.archive.ubuntu.com/ubuntu resolute/main amd64 whois amd64 5.6.6 [52.5 kB]
Fetched 510 kB in 1s (350 kB/s)
Selecting previously unselected package fail2ban.
(Reading database... 156747 files and directories currently installed.)
Preparing to unpack ./fail2ban_1.1.0-9_all.deb...
Unpacking fail2ban (1.1.0-9)...
Selecting previously unselected package python3-pyasyncore.
Preparing to unpack ./python3-pyasyncore_1.0.2-3build1_all.deb...
Unpacking python3-pyasyncore (1.0.2-3build1)...
Selecting previously unselected package python3-pyinotify.
Preparing to unpack ./python3-pyinotify_0.9.6-5build1_all.deb...
Unpacking python3-pyinotify (0.9.6-5build1)...
Selecting previously unselected package whois.
Preparing to unpack ./archives/whois_5.6.6_amd64.deb...
Unpacking whois (5.6.6)...
Setting up whois (5.6.6)...
Setting up fail2ban (1.1.0-9)...
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' → '/usr/lib/systemd/system/fail2ban.service'.
Setting up python3-pyasyncore (1.0.2-3build1)...
Setting up python3-pyinotify (0.9.6-5build1)...
Processing triggers for man-db (2.13.1-1build1)...
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-08-26 07:19:18 UTC; 4s ago
     Invocation: b8fb168f421c4b3cb2729d4f0b5c0254
       Docs: man:fail2ban(1)
    Main PID: 12222 (fail2ban-server)
       Tasks: 5 (limit: 1859)
      Memory: 15.9M (peak: 15.9M)
         CPU: 185ms
    CGroup: /system.slice/fail2ban.service
            └─12222 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

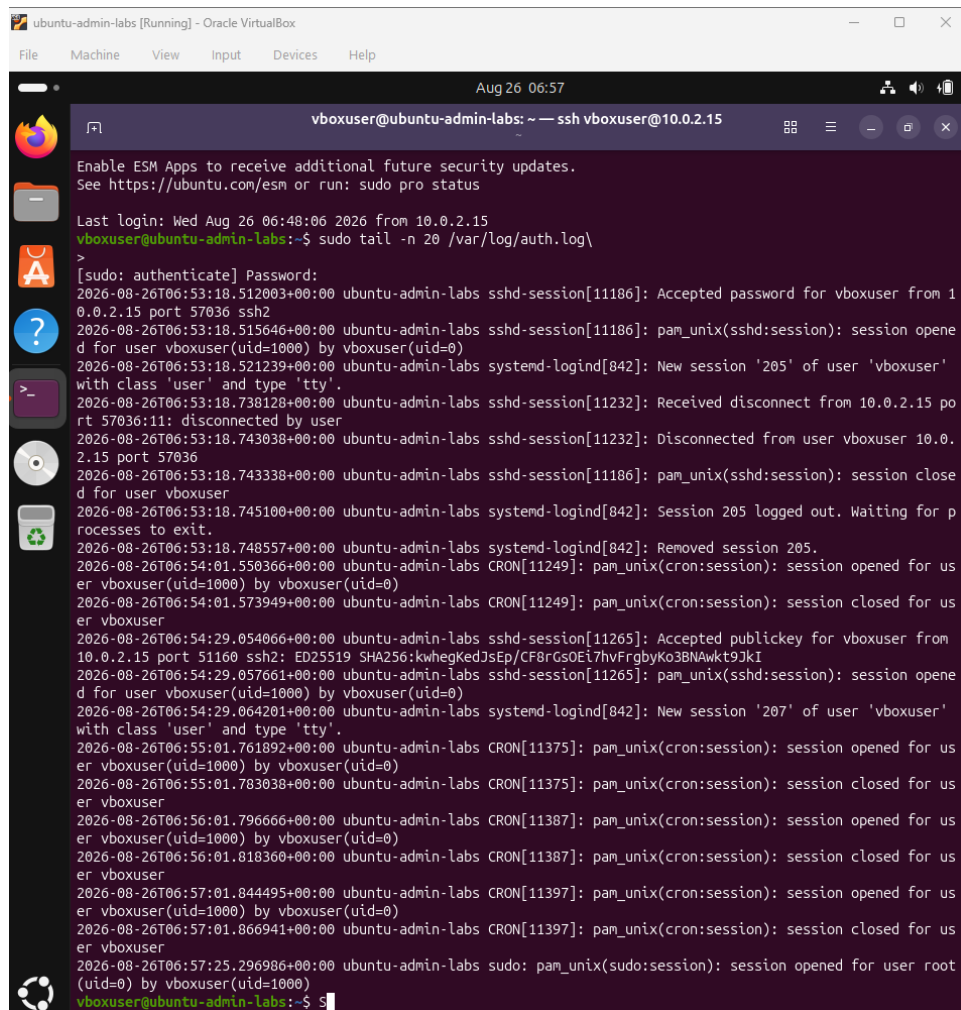
Aug 26 07:19:18 ubuntu-admin-labs systemd[1]: Started fail2ban.service - Fail2Ban Service.
Aug 26 07:19:18 ubuntu-admin-labs fail2ban-server[12222]: Server ready
```

SS/06-firewall-fail2ban-active.png — fail2ban installed and running, actively watching the system's authentication logs

7. Logs & Monitoring

This section covers reviewing the historical record of system activity: authentication events, resource usage, and logs scoped to a single service.

Command	Purpose
<code>sudo tail -n 20 /var/log/auth.log</code>	Review the most recent authentication events: logins, sudo usage, and SSH activity
<code>df -h</code>	Disk space usage per filesystem, in human-readable units
<code>free -h</code>	Memory usage, with the available column indicating what can actually still be used
<code>uptime</code>	System uptime, number of active sessions, and load average
<code>journalctl -u <service> --no-pager tail -n 20</code>	Recent log history for one specific systemd service



```
ubuntu-admin-labs [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Aug 26 06:57
vboxuser@ubuntu-admin-labs: ~ — ssh vboxuser@10.0.2.15

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Aug 26 06:48:06 2026 from 10.0.2.15
vboxuser@ubuntu-admin-labs:~$ sudo tail -n 20 /var/log/auth.log
>
[sudo: authenticate] Password:
2026-08-26T06:53:18.512003+00:00 ubuntu-admin-labs sshd-session[11186]: Accepted password for vboxuser from 10.0.2.15 port 57036 ssh2
2026-08-26T06:53:18.515646+00:00 ubuntu-admin-labs sshd-session[11186]: pam_unix(sshd:session): session opened for user vboxuser(uid=1000) by vboxuser(uid=0)
2026-08-26T06:53:18.521239+00:00 ubuntu-admin-labs systemd-logind[842]: New session '205' of user 'vboxuser' with class 'user' and type 'tty'.
2026-08-26T06:53:18.738128+00:00 ubuntu-admin-labs sshd-session[11232]: Received disconnect from 10.0.2.15 port 57036:11: disconnected by user
2026-08-26T06:53:18.743038+00:00 ubuntu-admin-labs sshd-session[11232]: Disconnected from user vboxuser 10.0.2.15 port 57036
2026-08-26T06:53:18.743338+00:00 ubuntu-admin-labs sshd-session[11186]: pam_unix(sshd:session): session closed for user vboxuser
2026-08-26T06:53:18.745100+00:00 ubuntu-admin-labs systemd-logind[842]: Session 205 logged out. Waiting for processes to exit.
2026-08-26T06:53:18.748557+00:00 ubuntu-admin-labs systemd-logind[842]: Removed session 205.
2026-08-26T06:54:01.550366+00:00 ubuntu-admin-labs CRON[11249]: pam_unix(cron:session): session opened for user vboxuser(uid=1000) by vboxuser(uid=0)
2026-08-26T06:54:01.573949+00:00 ubuntu-admin-labs CRON[11249]: pam_unix(cron:session): session closed for user vboxuser
2026-08-26T06:54:29.054066+00:00 ubuntu-admin-labs sshd-session[11265]: Accepted publickey for vboxuser from 10.0.2.15 port 51160 ssh2: ED25519 SHA256:kwhegKedJsEp/CFB8rCsOEi7hvFrgbyKo3BNAwkt9JkI
2026-08-26T06:54:29.057661+00:00 ubuntu-admin-labs sshd-session[11265]: pam_unix(sshd:session): session opened for user vboxuser(uid=1000) by vboxuser(uid=0)
2026-08-26T06:54:29.064201+00:00 ubuntu-admin-labs systemd-logind[842]: New session '207' of user 'vboxuser' with class 'user' and type 'tty'.
2026-08-26T06:55:01.761892+00:00 ubuntu-admin-labs CRON[11375]: pam_unix(cron:session): session opened for user vboxuser(uid=1000) by vboxuser(uid=0)
2026-08-26T06:55:01.783038+00:00 ubuntu-admin-labs CRON[11375]: pam_unix(cron:session): session closed for user vboxuser
2026-08-26T06:56:01.796666+00:00 ubuntu-admin-labs CRON[11387]: pam_unix(cron:session): session opened for user vboxuser(uid=1000) by vboxuser(uid=0)
2026-08-26T06:56:01.818360+00:00 ubuntu-admin-labs CRON[11387]: pam_unix(cron:session): session closed for user vboxuser
2026-08-26T06:57:01.844495+00:00 ubuntu-admin-labs CRON[11397]: pam_unix(cron:session): session opened for user vboxuser(uid=1000) by vboxuser(uid=0)
2026-08-26T06:57:01.866941+00:00 ubuntu-admin-labs CRON[11397]: pam_unix(cron:session): session closed for user vboxuser
2026-08-26T06:57:25.296986+00:00 ubuntu-admin-labs sudo: pam_unix(sudo:session): session opened for user root(uid=0) by vboxuser(uid=1000)
vboxuser@ubuntu-admin-labs:~$ s
```

SS/07-logs-auth-log-password-publickey.png — tail -n 20 on the authentication log, showing both a password-based login and a publickey-based login recorded with full detail

A screenshot of a terminal window titled 'vboxuser@ubuntu-admin-labs: ~ — ssh vboxuser@10.0.2.15'. The terminal shows the output of several commands: 'df -h' showing disk usage for various filesystems; 'free -h' showing memory usage; and 'journalctl -u ssh --no-pager | tail -n 20' showing a log of SSH sessions. The log includes failed password attempts and a successful publickey login for the 'vboxuser' at the end. The terminal window is part of an Oracle VM VirtualBox interface, with the host name 'ubuntu-admin-labs [Running]' visible at the top.

SS/07-logs-journalctl-ssh-history.png — journalctl -u ssh showing detailed SSH session history, including earlier failed password attempts and a later accepted publickey login

8. Bonus: Shell Scripting & Integration

This closing section covers writing and debugging a small bash script, and chaining several tools together to observe a firewall rule take effect in real time.

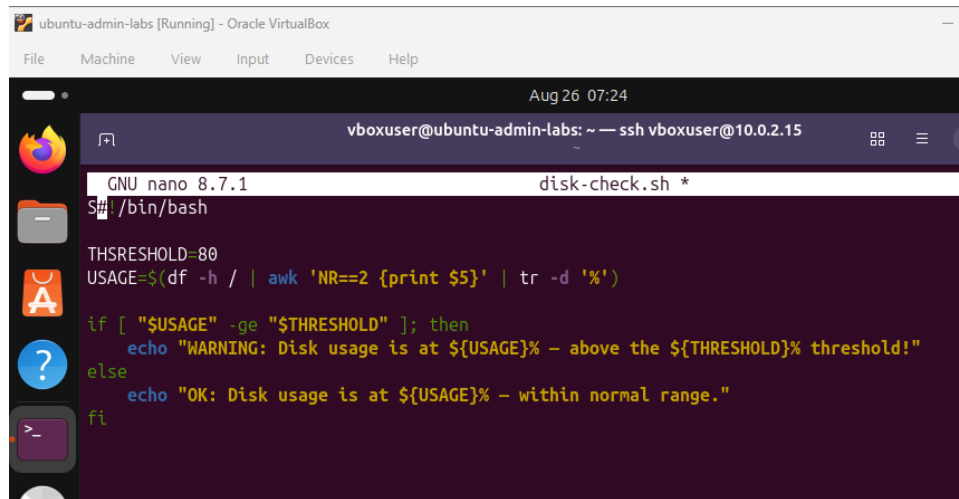
disk-check.sh

A script that checks root filesystem usage against a configurable threshold and prints a warning if it is exceeded. The finished script is included in this repository as disk-check.sh.

```
#!/bin/bash THRESHOLD=80 USAGE=$(df -h / | awk 'NR==2 {print $5}' | tr -d '%') if [
"$USAGE" -ge "$THRESHOLD" ]; then echo "WARNING: Disk usage is at ${USAGE}% – above the
${THRESHOLD}% threshold!" else echo "OK: Disk usage is at ${USAGE}% – within normal
range." fi
```

Debugging the script proved more instructive than writing it. Two issues were identified during development: a stray character had been introduced at the start of the shebang line, and the threshold variable had been misspelled on its assignment line (THSRESHOLD instead of THRESHOLD), meaning it was never actually set. Since the comparison further down correctly referenced the properly spelled variable, it silently evaluated against an empty

value. This was diagnosed by reviewing the script's actual saved contents with cat rather than assuming the intended text had been entered correctly — a useful general debugging practice.

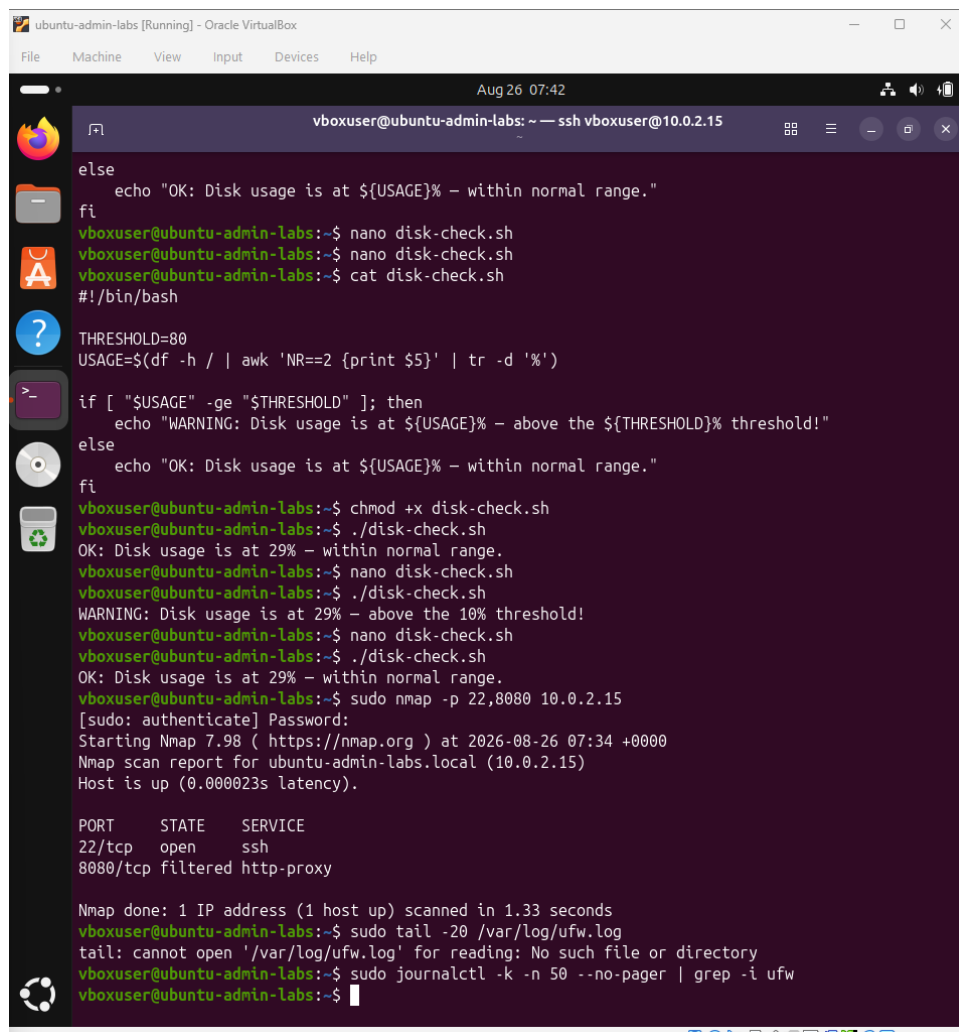


```
GNU nano 8.7.1 disk-check.sh *
#!/bin/bash

THRESHOLD=80
USAGE=$(df -h / | awk 'NR==2 {print $5}' | tr -d '%')

if [ "$USAGE" -ge "$THRESHOLD" ]; then
    echo "WARNING: Disk usage is at ${USAGE}% - above the ${THRESHOLD}% threshold!"
else
    echo "OK: Disk usage is at ${USAGE}% - within normal range."
fi
```

SS/08-bonus-script-bugs-and-fix.png — cat disk-check.sh revealing both bugs, followed by the corrected version after fixing the shebang and the variable name



```
else
    echo "OK: Disk usage is at ${USAGE}% - within normal range."
fi

vboxuser@ubuntu-admin-labs:~$ nano disk-check.sh
vboxuser@ubuntu-admin-labs:~$ nano disk-check.sh
vboxuser@ubuntu-admin-labs:~$ cat disk-check.sh
#!/bin/bash

THRESHOLD=80
USAGE=$(df -h / | awk 'NR==2 {print $5}' | tr -d '%')

if [ "$USAGE" -ge "$THRESHOLD" ]; then
    echo "WARNING: Disk usage is at ${USAGE}% - above the ${THRESHOLD}% threshold!"
else
    echo "OK: Disk usage is at ${USAGE}% - within normal range."
fi

vboxuser@ubuntu-admin-labs:~$ chmod +x disk-check.sh
vboxuser@ubuntu-admin-labs:~$ ./disk-check.sh
OK: Disk usage is at 29% - within normal range.
vboxuser@ubuntu-admin-labs:~$ nano disk-check.sh
vboxuser@ubuntu-admin-labs:~$ ./disk-check.sh
WARNING: Disk usage is at 29% - above the 10% threshold!
vboxuser@ubuntu-admin-labs:~$ nano disk-check.sh
vboxuser@ubuntu-admin-labs:~$ ./disk-check.sh
OK: Disk usage is at 29% - within normal range.
vboxuser@ubuntu-admin-labs:~$ sudo nmap -p 22,8080 10.0.2.15
[sudo: authenticate] Password:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-26 07:34 +0000
Nmap scan report for ubuntu-admin-labs.local (10.0.2.15)
Host is up (0.000023s latency).

PORT      STATE      SERVICE
22/tcp    open       ssh
8080/tcp   filtered  http-proxy

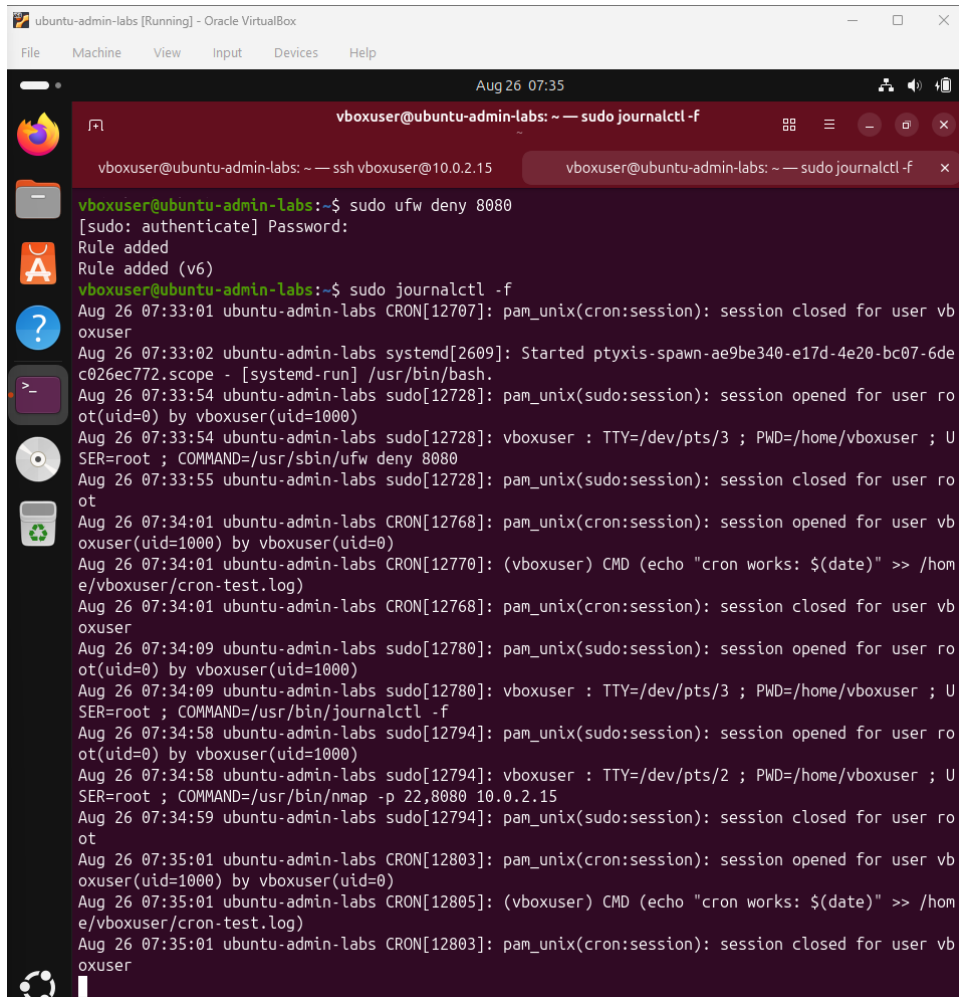
Nmap done: 1 IP address (1 host up) scanned in 1.33 seconds
vboxuser@ubuntu-admin-labs:~$ sudo tail -20 /var/log/ufw.log
tail: cannot open '/var/log/ufw.log' for reading: No such file or directory
vboxuser@ubuntu-admin-labs:~$ sudo journalctl -k -n 50 --no-pager | grep -i ufw
vboxuser@ubuntu-admin-labs:~$
```

SS/08-bonus-script-ok-warning-final-nmap.png — The corrected script producing both a genuine OK result and, with the threshold temporarily lowered, a genuine WARNING result

Integration exercise: ufw + nmap + journalctl

To directly observe a firewall change take effect rather than assume it based on configuration alone, three tools were used together:

- `sudo ufw deny 8080` — explicitly block a specific port
- `sudo journalctl -f` (in a second terminal) — follow the live system log
- `sudo nmap -p 22,8080 <host>` — scan both ports, confirming 22 remained open while 8080 changed to filtered



```
ubuntu-admin-labs [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-admin-labs: ~ — sudo journalctl -f
vboxuser@ubuntu-admin-labs: ~ — ssh vboxuser@10.0.2.15
vboxuser@ubuntu-admin-labs: ~ — sudo journalctl -f

vboxuser@ubuntu-admin-labs:~$ sudo ufw deny 8080
[sudo: authenticate] Password:
Rule added
Rule added (v6)
vboxuser@ubuntu-admin-labs:~$ sudo journalctl -f
Aug 26 07:33:01 ubuntu-admin-labs CRON[12707]: pam_unix(cron:session): session closed for user vb
oxuser
Aug 26 07:33:02 ubuntu-admin-labs systemd[2609]: Started ptyxis-spawn-ae9be340-e17d-4e20-bc07-6de
c026ec772.scope - [systemd-run] /usr/bin/bash.
Aug 26 07:33:54 ubuntu-admin-labs sudo[12728]: pam_unix(sudo:session): session opened for user ro
ot(uid=0) by vboxuser(uid=1000)
Aug 26 07:33:54 ubuntu-admin-labs sudo[12728]: vboxuser : TTY=/dev/pts/3 ; PWD=/home/vboxuser ; U
SER=root ; COMMAND=/usr/sbin/ufw deny 8080
Aug 26 07:33:55 ubuntu-admin-labs sudo[12728]: pam_unix(sudo:session): session closed for user ro
ot
Aug 26 07:34:01 ubuntu-admin-labs CRON[12768]: pam_unix(cron:session): session opened for user vb
oxuser(uid=1000) by vboxuser(uid=0)
Aug 26 07:34:01 ubuntu-admin-labs CRON[12770]: (vboxuser) CMD (echo "cron works: $(date)" >> /hom
e/vboxuser/cron-test.log)
Aug 26 07:34:01 ubuntu-admin-labs CRON[12768]: pam_unix(cron:session): session closed for user vb
oxuser
Aug 26 07:34:09 ubuntu-admin-labs sudo[12780]: pam_unix(sudo:session): session opened for user ro
ot(uid=0) by vboxuser(uid=1000)
Aug 26 07:34:09 ubuntu-admin-labs sudo[12780]: vboxuser : TTY=/dev/pts/3 ; PWD=/home/vboxuser ; U
SER=root ; COMMAND=/usr/bin/journalctl -f
Aug 26 07:34:58 ubuntu-admin-labs sudo[12794]: pam_unix(sudo:session): session opened for user ro
ot(uid=0) by vboxuser(uid=1000)
Aug 26 07:34:58 ubuntu-admin-labs sudo[12794]: vboxuser : TTY=/dev/pts/2 ; PWD=/home/vboxuser ; U
SER=root ; COMMAND=/usr/bin/nmap -p 22,8080 10.0.2.15
Aug 26 07:34:59 ubuntu-admin-labs sudo[12794]: pam_unix(sudo:session): session closed for user ro
ot
Aug 26 07:35:01 ubuntu-admin-labs CRON[12803]: pam_unix(cron:session): session opened for user vb
oxuser(uid=1000) by vboxuser(uid=0)
Aug 26 07:35:01 ubuntu-admin-labs CRON[12805]: (vboxuser) CMD (echo "cron works: $(date)" >> /hom
e/vboxuser/cron-test.log)
Aug 26 07:35:01 ubuntu-admin-labs CRON[12803]: pam_unix(cron:session): session closed for user vb
oxuser
```

SS/08-bonus-ufw-deny-journalctl-live.png — ufw deny 8080 and journalctl -f, capturing the live, timestamped audit trail of both the firewall change and the nmap scan

Notable Troubleshooting

The following issues were encountered and resolved over the course of this project:

- VirtualBox Guest Additions failed to run when executed directly, because its installer CD is mounted read-only via ISO9660 with the execute permission removed, even for root. Resolved by running the installer through the shell interpreter instead: `sudo bash VBoxLinuxAdditions.run`.
- `sudo reboot` was blocked by a systemd inhibitor protecting the active desktop session. Resolved with `sudo systemctl reboot -i`, where `-i` explicitly ignores inhibitors.
- A saved crontab initially had no effect because a stray character had been accidentally left at the top of the file during editing. Resolved by removing the bad line in nano before saving.

- disk-check.sh's threshold comparison failed with an "integer expected" error, ultimately traced to a misspelled variable name rather than a logic error, as detailed in the bonus section above — resolved by reading the file's actual saved content rather than assuming the intended text had been entered correctly.
-

Key Takeaways

- Most of the practical work in this lab involved careful interpretation of command output — the results of ss, nmap, and log files — rather than memorization of syntax.
- The majority of issues encountered were the result of typographical errors or incorrect sequencing, such as enabling the firewall before allowing SSH, rather than gaps in conceptual understanding.
- The same tools used to inspect system state (ps, ss, journalctl) were also used to verify that configuration changes had taken effect, through direct before-and-after comparison.